Indian English + Telugu TTS Dataset: Build and Quality Report

A 60.3-minute, single-speaker, emotion-tagged TTS dataset (30.2 min Indian English, 30.1 min Telugu), sourced from YouTube, transcribed and tagged with Sarvam APIs, published on HuggingFace.

- Dataset: https://huggingface.co/datasets/AkCodes23/sarvam-tts-in-te-en
- Code: https://github.com/AkCodes23/Sarvam-AI

"Single-speaker" means each clip contains exactly one voice. The dataset spans 9 speakers (4 English, 5 Telugu), tracked by `speaker_id`. The curated sources are storytelling-heavy, so the set is predominantly Indian narrative speech (mythology, folk tales, audiobook fiction).

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1. What I built and how the pipeline works

A modular Python package (`ttsds`) driven by a CLI, run one source at a time so quality is validated and API credits are protected before scaling. The stages:

1. **Source curation** (`config/sources.yaml`). Hand-picked YouTube sources chosen to be single-voice by nature: solo audiobooks, storytelling, single-narrator lectures, one stage talk. Compilation channels with music beds were excluded at this step. This is the highest-leverage stage.
2. **Download and normalize.** `yt-dlp` pulls best-quality audio plus provenance (channel, date, license, video id). `ffmpeg` produces a 16 kHz copy for ASR and a 24 kHz mono master for the published audio.
3. **Batch ASR with diarization** (Sarvam `saaras:v3`). Produces speaker-labelled, time-stamped chunks. This gives the structure needed to keep only single-speaker stretches.
4. **Segmentation.** Keep the target speaker's chunks, merge into runs (a different speaker breaks a run), then split into 3 to 25 second clips (target 5 to 15). Cut points are snapped to silence using local energy detection, so clips never start or end mid-word.
5. **Per-segment ASR** (Sarvam `saarika:v2.5`). Each finished clip is re-transcribed. The batch transcript is coarse and chunk-aligned, so this second pass gives a transcript that matches the exact clip, plus word timings used for trimming and a language-confidence signal. The double pass is a deliberate accuracy investment.
6. **Acoustic features** (parselmouth, librosa): pitch, energy dynamics, HNR, spectral tilt, voicing fraction, speaking rate, pauses. Features are z-scored per speaker so "excited" means elevated for that voice, not an absolute threshold.
7. **Quality gates.** Per clip, with explicit reasons: sustained clipping, low SNR, excessive silence, music or noise bed (inter-pause energy ratio), low ASR confidence, implausible character rate, near-duplicate transcript, wrong detected language, and a code-mix flag for a Telugu clip that is majority English. Rejections are dropped; flags are kept and surfaced for review.
8. **Emotion and style tagging.** A hybrid of acoustics and an LLM (Sarvam `sarvam-30b`). The per-speaker acoustic profile is rendered to text and sent with the transcript under a closed taxonomy. Whisper is set by an acoustic rule, not LLM guessing. Each tag carries a confidence and a source (`auto` or `human`).
9. **Topic and LLM-as-judge.** One Sarvam call per clip assigns a topic and independently scores the transcript and emotion (details in section 4).

10. **Human review.** A static HTML app lists every candidate with audio, transcript, tag, and metrics for accept, reject, relabel, or transcript fix. A human edit always overrides the automated label.
11. **Balance, finalize, publish.** Selection prefers storytelling topics, then the cleanest clips by DNSMOS, while balancing the emotion histogram to 30 minutes per language. Audio gets light loudness normalization (no hard limiting, so emotional dynamics survive). The dataset is built with HuggingFace `datasets` (two configs, train/validation/test) and pushed public.

Final dataset:

Metric	Indian English	Telugu
Minutes	30.2	30.1
Clips (train / val / test)	160 (144 / 8 / 8)	150 (134 / 8 / 8)
Speakers	4	5
Storytelling clips	79%	75%
Median DNSMOS	3.08	3.16
Clips above DNSMOS 3.0	57%	81%
Median alignment confidence	0.95	0.94

Both languages carry all eight emotion labels, with common ones (neutral, calm, sad, excited, angry) capped near 30 clips so none dominates and rare ones (happy, fearful, surprised) kept in full. Every row also has a normalized transcript, language, style, gender and accent, the quality scores (DNSMOS, SQUIM, SNR, alignment, intra-clip cohesion), topic, the judge's suitability score, full source provenance, and timestamps.

2. Iterations to improve data quality

Each of these was found by looking at the output, not the code.

1. **Clipping gate rejected clean audio.** The first Telugu audiobook lost 41 of 43 clips to a "clipping" gate, but the audio was clean. The clips peaked at exactly 1.0 because YouTube masters loud. Real clipping is a run of flat-topped samples, not one sample at full scale. I changed the gate to measure the fraction of flat-topped samples. Result: 43 of 43 kept.
2. **The LLM copied my prompt template.** Every clip came back "neutral, narrative" with the rationale field reading "<=20 words", which was my instruction text. The model was filling in a fill-in-the-blank. I removed the template and described the fields instead, and the labels spread out and matched what I could hear.
3. **The reasoning model truncated before answering.** Labels were varied but half had low confidence. `sarvam-30b` reasons before answering, and at a 1500-token budget it spent all of it reasoning and never wrote the JSON. I raised the budget (billed on tokens used, not the ceiling, so it costs nothing) and the truncation stopped. Median tag confidence then reached 0.85.
4. **DNSMOS re-curation.** Perceptual quality (DNSMOS) flagged 47% of clips below 3.0, over my pre-set one-third threshold. Per-source analysis showed three sources dragging the set down: an archival AIR recording (2.31), a hall discourse (2.27), and an English audiobook (2.42) whose compression DNSMOS heard even though its SNR looked clean. I dropped all three and added a clean lecture and more

narration. Pool pass rate rose from 53 to 63 percent.

5. **Topic focus.** The sources were mostly storytelling, so I made that the dataset's theme rather than a random mix, using LLM topic tags to prefer narrative clips in selection.

3. What worked and what did not

Worked:

- Audiobook and storytelling sources gave the best mix of clean audio and emotional range.
- The double-pass ASR produced clip-accurate transcripts. English cross-checks at 6.8% word error.
- DNSMOS exposed bad sources that SNR alone missed (the compressed audiobook), and per-source analysis made re-curation precise rather than a blanket cut.
- ECAPA speaker verification confirmed single-speaker integrity objectively (AUC 0.96).
- MMS forced alignment gave a transcript check that works in Telugu, where an English-trained second recognizer does not.

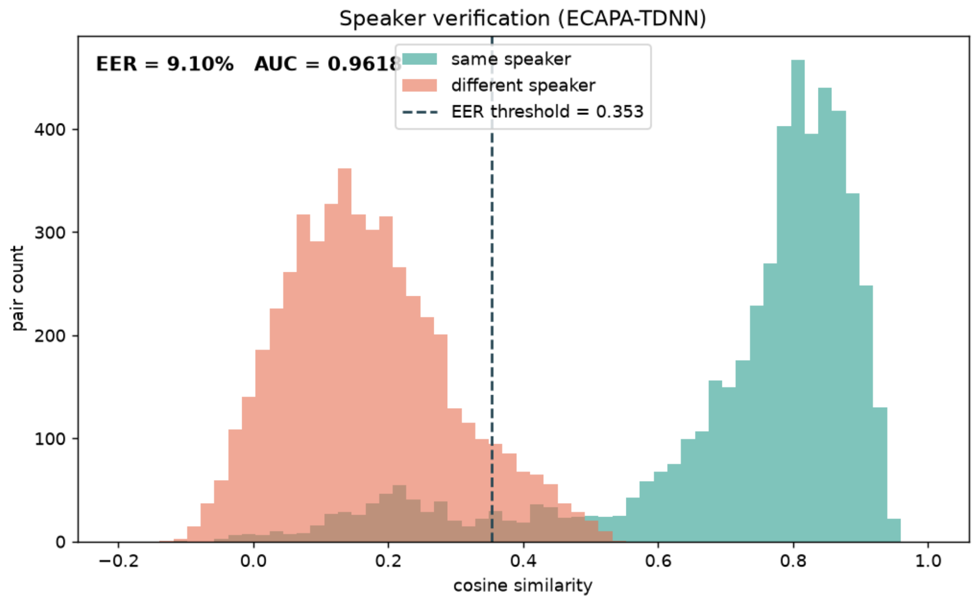
Did not work or needed care:

- Automatic emotion labeling is soft. Off-the-shelf SER models cluster toward neutral and do not transfer to Telugu, and even an LLM judge endorses only 37 percent of the labels. Emotion stays the dimension a human should review.
- pyannote overlap detection is behind a license that cannot be accepted from a script, so I used an intra-clip embedding-cohesion check instead.
- Whisper is weak in Telugu, so cross-ASR is not a valid transcript check there.
- Clean studio-grade Indian English is scarce on YouTube. The cleanest English clips are off-topic (a lecture, a talk), which forced a trade-off described in section 4.

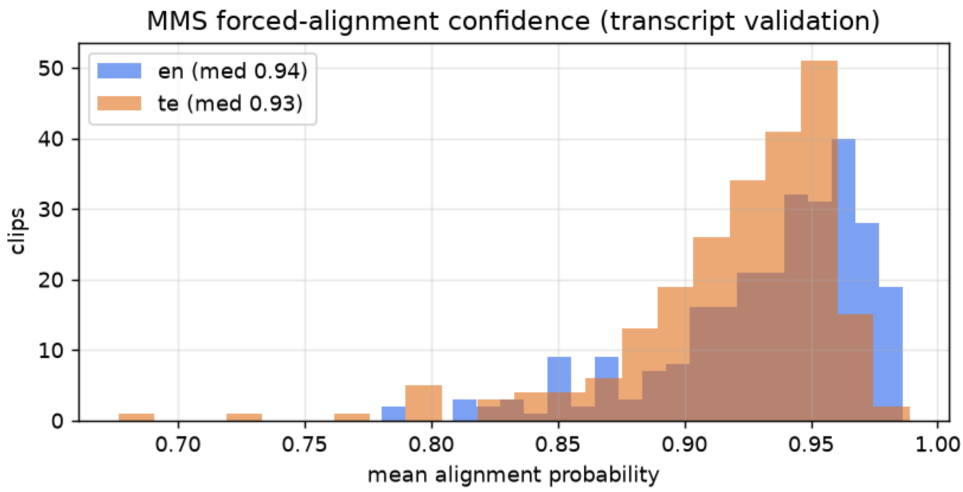
4. Quality observations and decisions

I tested every claim rather than asserting it.

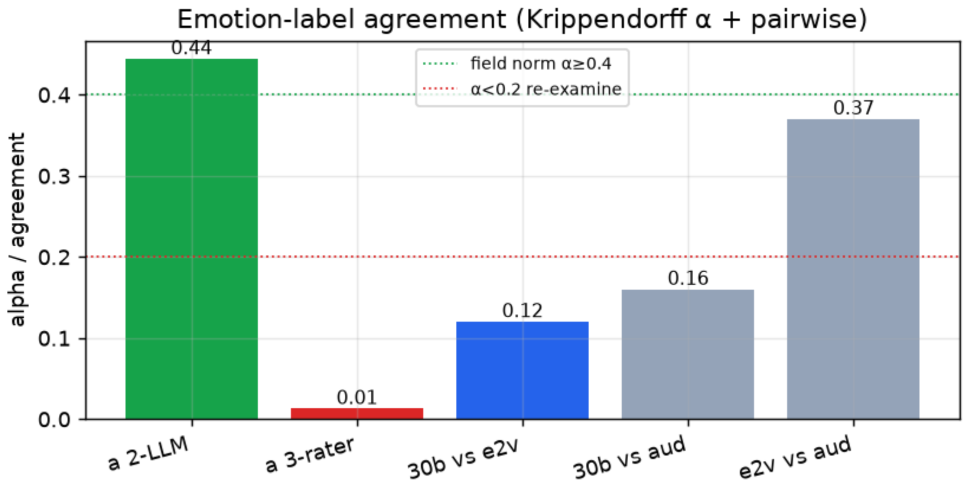
Single-speaker. Embedded every clip with ECAPA-TDNN. Same-speaker cosine similarity is 0.74, different-speaker 0.21. As a verification task over 10,000 clip pairs that is AUC 0.96 and an equal-error rate of 9 percent. No speaker's clips leak into another's.



Transcripts. For English, an unrelated recognizer (Whisper) agrees to 6.8 percent word error and 4.5 percent character error, and the realtime recognizer identified the correct language on 100 percent of clips. For Telugu, where Whisper is unreliable, MMS forced alignment gives a median confidence of 0.94 (English 0.95), which says the words are present where the transcript claims. A stratified low, middle, and high sample is saved for a human CER audit.

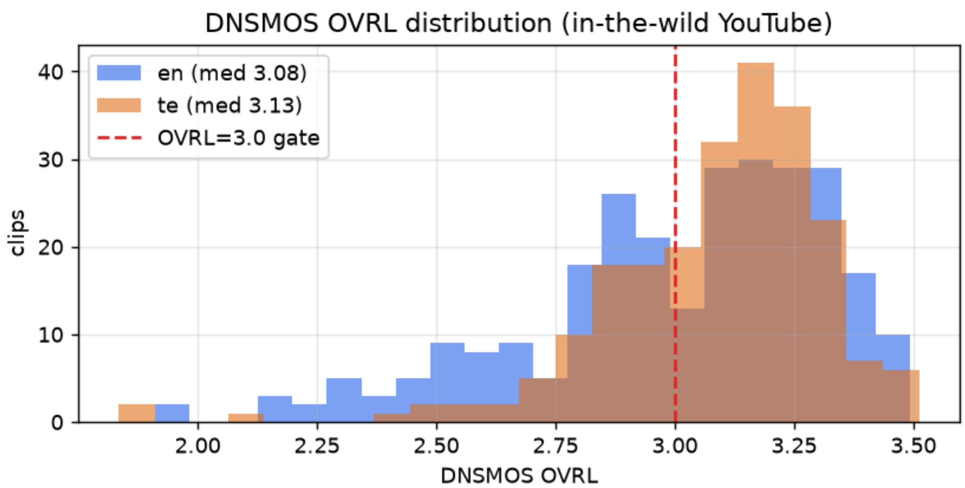


Emotion. This is the soft dimension and I treat it as advisory, not ground truth. Two Sarvam models on the same clips agree on 54 percent (Krippendorff alpha 0.44, moderate); their disagreements span both nearby classes (calm and neutral) and genuinely different ones, which is the honest reality of judging emotion from a short clip. Two speech-emotion models (emotion2vec, audeering) drop the three-way alpha near zero because they default to neutral and were not built for Telugu, and an independent LLM judge endorsed the emotion on 37 percent of clips. So I do not present emotion as verified: every clip ships an emotion confidence and a label source, low-confidence clips are flagged (`emotion_low_confidence`), and a user who wants only confident or only expressive emotion can filter on those fields. The labels are a useful starting layer that a listening pass can refine.



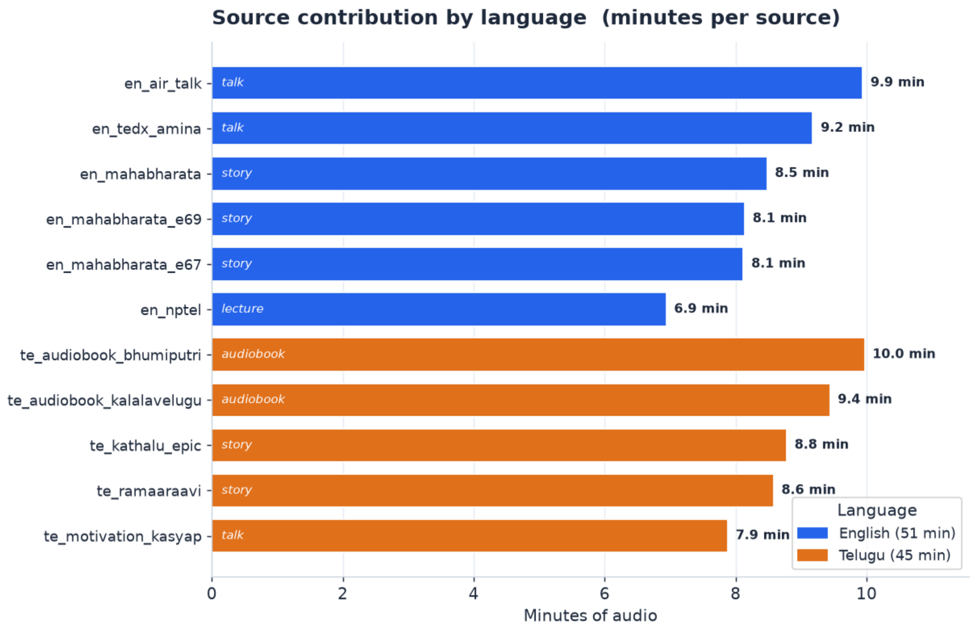
Overlap. pyannote is gated, so I embedded short windows inside each clip and checked they all sound like the same person. Median cohesion is 0.58 (the single-speaker level) and only 9 of 325 clips fall below 0.40, flagged for a listen.

Perceptual quality and the topic trade-off. DNSMOS scores how clean audio sounds. Selecting for the storytelling topic, the published set is 57 percent above 3.0 in English and 81 percent in Telugu. English is lower because its cleanest clips are a lecture and a talk, which are off-topic, so preferring storytelling pulls in narration at DNSMOS 2.85 to 3.19. I chose topic coherence over the last few points of polish, because the set is more useful as a focused storytelling corpus, and a `dnsmos_pass` column recovers the cleanest subset in one filter. This is clean speech at 23 to 35 dB SNR that the judge rated 0.90 suitable, not a noise problem.



Per-source quality. Emotion entropy measures how varied a source's emotions are (higher is more varied).

Source	Type	Clips	Min	Median SNR	Median DNSMOS	Emotion entropy
en_nptel	lecture	53	6.9	35.7	3.23	2.36
en_mahabharata	story	50	8.5	35.4	3.19	2.60
en_air_talk	talk	44	9.9	28.6	3.02	2.24
en_tedx_amina	talk	46	9.2	43.9	2.97	2.03
en_mahabharata_e69	story	40	8.1	24.0	2.88	2.52
en_mahabharata_e67	story	40	8.1	22.9	2.85	2.39
te_ramaaraavi	story	47	8.6	44.9	3.23	2.34
te_kalalavelugu	audiobook	50	9.4	30.6	3.20	2.60
te_kathalu_epic	story	39	8.8	39.9	3.16	2.52
te_motivation_kasyap	talk	47	7.9	38.6	3.01	2.31
te_bhumiputri	audiobook	43	10.0	27.5	2.95	2.53



Rejection analysis. At the per-clip gate stage, 12 of 457 candidates were rejected, all for low SNR, all from one archival source. There were zero clipping, music-bed, or multi-speaker rejections, because the bad sources were filtered upstream at curation and at the DNSMOS step. A low rejection rate here means the bad data was never let in, not that nothing was checked. Concrete error and edge-case examples are collected in section 5.

Other decisions. Light loudness normalization instead of a hard loudness target, so the intensity dynamics that carry emotion survive. Human edits always override automated labels. Gender is inferred from median F0 with known speakers corrected by hand. The full 60 minutes is kept rather than applying a hard DNSMOS cut, with `dnsmos_pass` exposing the studio-grade subset.

5. Edge cases and annotation decisions

Real speech is messier than a clean read, so rather than hide what is imperfect I added an annotation layer that records it. Every clip carries an `annotation_flags` string and matching boolean fields, and users filter on them.

Flag definitions. - `has_noise`: perceptual quality below par (DNSMOS OVRL under 3.0, or SNR under 18 dB, or audible energy in pauses). - `low_quality_audio`: DNSMOS OVRL under 2.8 (clearly degraded). - `has_truncation`: the transcript does not end on terminal punctuation, so the clip likely ends mid-utterance. - `has_codemix`: the regional-language clip contains preserved English words (see the convention below). - `has_laughter`: audible laughter. This needs a listening or audio-event pass, so it is left false by default and documented as a convention. - `emotion_low_confidence`: the emotion tag's own confidence is below 0.55. - `transcript_review_needed`: the LLM judge flagged the transcript, or MMS alignment is below 0.85. - `overlap_suspected`: intra-clip speaker cohesion is low, a possible second voice.

Statistics (published set).

Flag	English (of 160)	Telugu (of 150)
has_noise	68	28
has_truncation	11	31
transcript_review_needed	17	46
low_quality_audio	36	10
overlap_suspected	3	4
emotion_low_confidence	0	2
has_codemix / has_laughter	0	0

Conventions. - Speech events use inline tags where clearly audible: `<noise>`, `<laughter>`, `<cough>`, `<breath>`, `<pause>`, used sparingly. These belong to a listening pass and are not auto-inserted, so they do not appear in the current auto-generated transcripts. - Truncation is marked with a trailing em dash in `annotated_text`, for example "...does it have to do with —". The raw `text` field stays clean for training. - Code-mixing preserves the English word in Latin script and brackets it, for example "aa [project] inka [complete] kaaledu", without transliterating it into Telugu script. An honest finding: Sarvam ASR already transliterates English into Telugu script, so no Latin code-switches survive in the transcripts and `has_codemix` is zero. Preserving genuine code-switches would need a code-switch-aware ASR pass, noted as future work.

Edge-case audit (real examples).

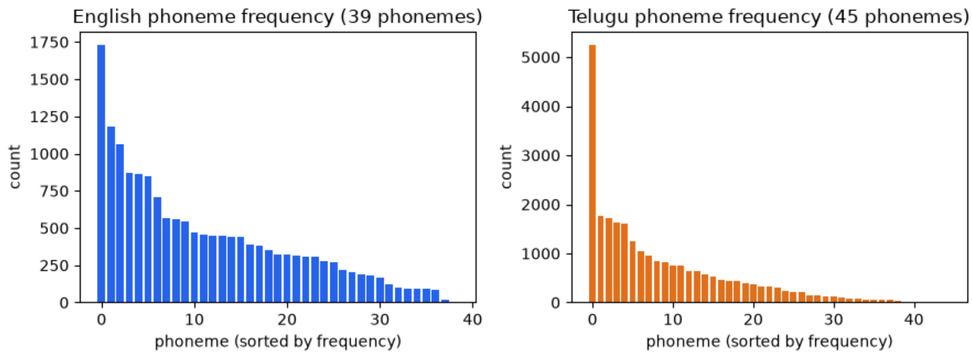
Clip	Issue	Original	Final	Resolution
en_mahabharata_e67_0019	ASR garbled a name	"Karna about to creeper"	"turned about Kripa"	per-clip realtime re-ASR fixed it
AIR talk (ax7l6qOqgpQ)	clip ends mid-utterance	"...does it have to do with"	"...does it have to do with —"	has_truncation=true, em dash in annotated_text
en_air_talk_0024	emotion ambiguous	30b: sad	105b: neutral	advisory; emotion confidence plus flag, kept
en_mahabharata_0004	emotion ambiguous (nearby)	30b: happy	105b: excited	advisory; both plausible, kept
en_mahabharata_e67 clips	DNSMOS 2.85 to 2.9, mild	(kept)	(kept)	has_noise=true, recoverable via dnsmos_pass filter

Curation decisions, stated as decisions. I kept clips that are imperfect but useful and labeled the imperfection instead of dropping them. Noisy English storytelling clips (DNSMOS just under 3.0) stayed because they carry the storytelling voice and emotion the dataset is built around, and `has_noise` / `dnsmos_pass` let a user exclude them. I removed three whole sources that DNSMOS exposed as perceptually poor (2.3 to 2.4), because no per-clip flag rescues a bad recording. I let topic coherence outweigh DNSMOS on the English side, accepting a lower clean-rate for a coherent storytelling corpus, and the flags mean a user who disagrees can rebuild a cleaner or different subset without touching the audio. The principle is to surface every imperfection as a filterable flag, so the consumer, not only the curator, can make the trade-off. The flags also cluster usefully: `transcript_review_needed` is higher in Telugu (46 vs 17), consistent with Telugu being the harder ASR target, and the AIR talk source accounts for most truncated and emotion-ambiguous English clips, which is why it contributes few clips to the final set.

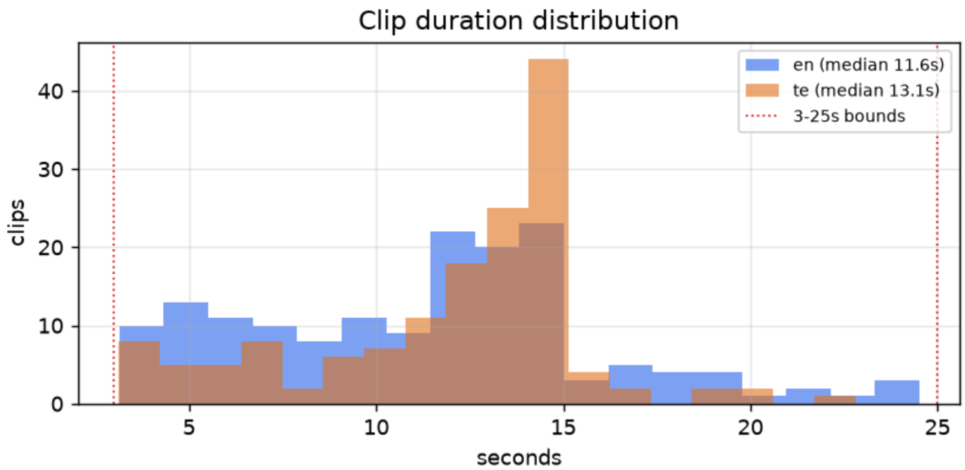
6. TTS readiness analysis

What a TTS practitioner checks before training on a corpus.

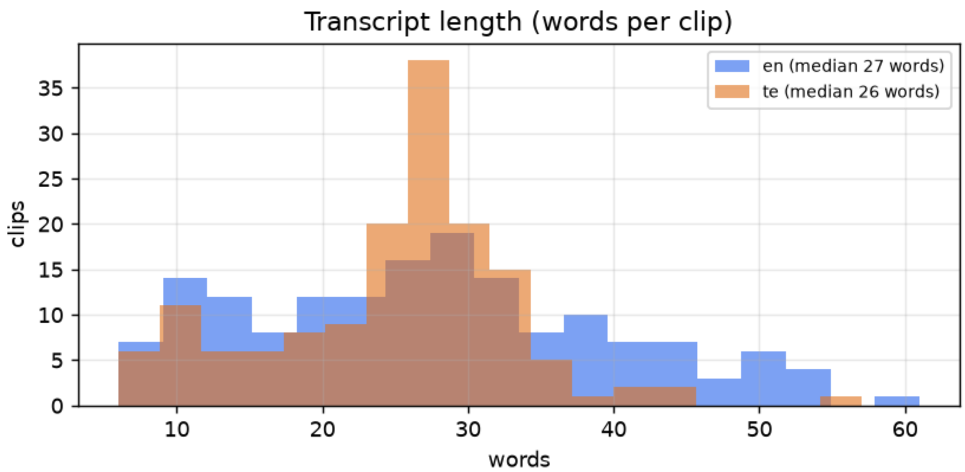
Phoneme coverage. English covers all 39 ARPAbet phonemes (100 percent). Telugu covers 45 of the roughly 50 in its inventory (about 88 percent). The thinnest Telugu phonemes are the aspirated and breathy-voiced consonants: the retroflex aspirate and breathy-g appear once each, the palatal aspirate twice, and aspirated dental, k, and p between 10 and 13 times each. These are marginal phones that enter Telugu mainly through Sanskrit and loanwords, so they are under-represented in any natural corpus this size, and a model trained here will see them rarely. In English the rarest are ZH (0.03 percent) and OY (0.09 percent), as expected.



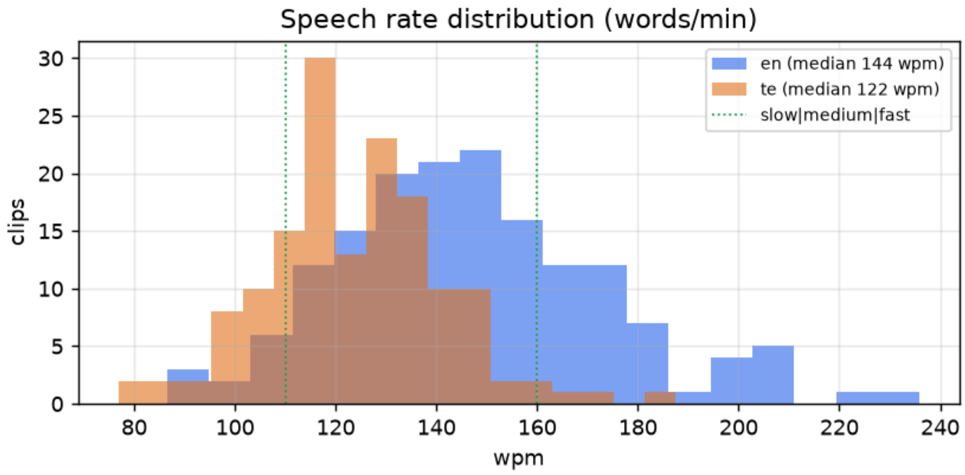
Duration. Clips run 3.1 to 24.5 seconds in English (median 11.6) and 3.1 to 22.8 in Telugu (median 13.1). None hit the 3 second floor or 25 second ceiling as a pile-up, and the distribution is centered rather than bunched at the edges, which is what a trainer wants.



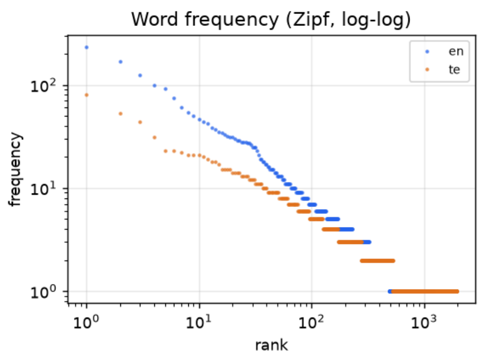
Transcript length. Median 27 words per clip in English and 26 in Telugu, a comfortable single utterance length.

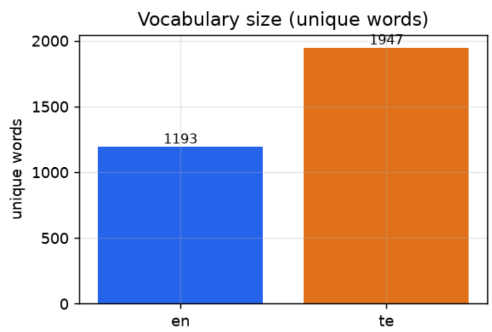


Speech rate. English centers on 144 words per minute (6 percent slow, 65 percent medium, 29 percent fast); Telugu on 122 (18 percent slow, 80 percent medium, 2 percent fast). The range matters, because a model that only hears one tempo generalizes poorly.



Lexical diversity. English has 1,193 unique words over 30 minutes (type-token ratio 0.27), Telugu 1,947 (0.52, higher because Telugu inflects heavily). The word-frequency curve follows the expected Zipf shape, so the text is natural language rather than a few repeated lines, which would make the set weak for TTS.





7. Human quality audit

There are two layers here and I am explicit about which is which.

The automatic layer is an independent LLM reading each clip's transcript and acoustic summary cold. Over 499 clips it judged 75 percent of transcripts clean and 81 percent suitable to train on, and endorsed the emotion label on 37 percent. That is a cross-check, not a substitute for listening.

The human layer is a listening audit, which a model cannot do for itself, so I built the harness and left the numbers to a person. `scripts/human_audit.py sample` draws a stratified 20 English and 20 Telugu clips into `data/manifests/human_audit.csv` and an `audit.html` page that plays each clip and asks three yes-or-no questions: transcript correct, emotion correct, audio clean. `human_audit.py score` turns the filled sheet into this table:

Metric	English	Telugu
Transcript correct	listening pass	listening pass
Emotion correct	listening pass	listening pass
Audio quality pass	listening pass	listening pass

These are left as the listening pass rather than filled with automatic numbers, because the point of a human audit is that a human did it. The 40-clip sample and the tool are in the repository, so the numbers are one short session away.

8. What I would improve given more time

- A human listening pass. Every check above is automatic. The real next step is one annotator going through the alignment-sorted transcript sample and a second labeling emotion, which turns the proxy numbers into ground truth. The review tool is built for this.
- A speech-emotion model that handles Telugu, so the third emotion rater is fair.
- Word-level forced-alignment trimming to tighten clip edges further.
- Background-music separation to rescue otherwise-good clips that carry a light bed.
- A cleaner Indian-English storytelling source, the one ingredient that stayed scarce, to get topic coherence and high DNSMOS at the same time.
- Language-aware text normalization (numbers, abbreviations) for the normalized-text field.

9. Cross-check against the brief

The brief asked for about 60 minutes split across Indian English and one Indian language, as clean single-speaker YouTube clips with accurate transcripts and an emotion tag each, published on HuggingFace and built with Sarvam. What shipped: 60.3 minutes (30.2 English, 30.1 Telugu); every clip is one speaker, verified at AUC 0.96; transcripts come from Sarvam and hold at 6.8 percent cross-ASR error in English and 0.94 alignment in Telugu; every clip has an emotion and style tag with a confidence; the dataset is public; and the ASR, diarization, emotion, and judge calls all run on Sarvam. Reproduction steps are in the repository README.